

Machine Learning Developer Intern Technical Assessment

Candidate: Hamza Madi

Position: Machine Learning Developer Intern

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Overview

This document provides a summary of the completed technical assessment for the Machine Learning Developer Intern position.

The task involved performing **Exploratory Data Analysis (EDA)** on a Housing Prices dataset to uncover insights, identify trends, and prepare the data for potential modelling.

The assessment was implemented in **Python** using **Jupyter Notebook**, with libraries such as `pandas`, `numpy`, `matplotlib`, and `seaborn`.

Contents

1. **Jupyter Notebook (`HamzaMadi_ML_Assessment.ipynb`)**

Contains all steps of the assessment, including:

- Dataset acquisition and description
- Data understanding (structure, data types, missing values, duplicates)
- Data cleaning and preparation
- Exploratory Data Analysis (EDA) with descriptive statistics and visualizations
- Insights and interpretation of factors influencing house prices
- Optional feature engineering and preliminary modelling

2. **README.pdf** (this document)

Provides overview, setup instructions, notes, and conclusion.

Notes & Assumptions

- All categorical variables have been encoded appropriately where necessary.
 - Missing data has been handled using standard imputation or removal methods depending on context.
 - Data visualizations include distributions, correlations, and relationships with the target variable `SalePrice`.
 - Optional modelling steps (feature engineering and simple regression) are included but not required for the core assessment.
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Conclusion

This assessment demonstrates:

- Understanding of **data acquisition, cleaning, and preparation**
- Ability to perform **exploratory data analysis and visualization**
- Skills in **deriving actionable insights from data**
- Proper **notebook structure, readability, and documentation**